Lesson plan · for the teacher

Reading data safely

Data science · Level 1 · Lesson 4

Time
40 min

Materials
A claim from a newspaper, advert or poster; Paper for group answers; A phone or tablet with Darsi, for the lab

Learning goals

- Look inside groups before comparing averages.
- Ask what else could explain a pattern.
- Share results without exposing a child.

Activity: What else could explain it?

1. Warm-up (5 min): Read a claim such as “Pupils who sit at the front get better marks.” Ask: should we move everyone to the front?
2. Teach (8 min): Tell the morning and afternoon story, then the breakfast example. Model the question “What else could explain it?”
3. Check (15 min): Each group gets a claim card and writes two other explanations and the data that would check the claim.
4. Privacy (7 min): Show four ways of sharing results (names on the wall, the class average, a file with numbers, a ranked list). Pupils sort them into safe and unsafe.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Discussion

- What could explain the front-row pattern, apart from the seat?
- Why is a class of eight risky to judge on one test?
- How will you share the next test’s results?

Exit ticket

- Write one pattern from school life and one other thing that could explain it.

Try it in class tomorrow
Bring one claim from a newspaper or a poster. Ask your pupils what else could explain it, and what data would check it.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **What is the safest way to show parents how the class did?**

- ◯ Every pupil's mark, by name
- ◯ A ranked list from best to worst
- ◯ The class average and spread

2 **If the morning class has a higher average, the time of day must be the reason.**

- ◯ True
- ◯ False

3 **Each pair is (pupil number, mark). What does this program print, and is it safe to share?**

```
marks = [("P01", 14), ("P02", 9), ("P03", 17), ("P04", 12)]
total = 0
for pupil, mark in marks:
    total = total + mark
print(round(total / len(marks), 1))
```

- ◯ 52: the total, so it is safe to share
- ◯ Each pupil's number and mark, then 13.0: not safe to share
- ◯ 13.0 four times, once for each pupil
- ◯ 13.0: only the class mean, so it is safe to share

4 **Before acting on "the morning class does better", put these checks in a sensible order.**

- ☐ Compare similar pupils across the classes
- ☐ Ask who is in each class
- ☐ Only then decide what to change
- ☐ Look at more than one test

5 **Which habits keep pupils' data safe? Choose all that apply.**

- ☐ Use pupil numbers instead of names
- ☐ Collect only what the question needs
- ☐ Pin a ranked list on the wall
- ☐ Delete the data when you're done

Answers · for the teacher

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1 **The class average and spread**

The class picture tells parents enough without pointing at any child.

2 **False**

Look inside the groups first. Here, the afternoon class had most of the new pupils; within each group, both classes scored the same.

3 **13.0: only the class mean, so it is safe to share**

The loop only adds the marks (52); nothing is printed inside it.

4 **Top to bottom: 2, 1, 4, 3**

Look inside the groups first, compare like with like, check that the pattern holds over time, and decide last.

5 **Use pupil numbers instead of names; Collect only what the question needs; Delete the data when you're done**

Numbers instead of names, collecting less and deleting on time all protect pupils.

✎ **For you to think about**

Where does your school share pupils' results today? Is any child exposed?